

ADMN 873

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Homework Assignment #5

Introduction

I decided to investigate tweets that mentioned Aaron Rodgers. He is the most valuable potential player in this offseason if he decides to leave Green Bay. I was curious to see what the twitter world would be talking about relating to him, so I dug a bit deeper and here are my findings:

Process of Downloading, Cleaning and Analyzing Tweets

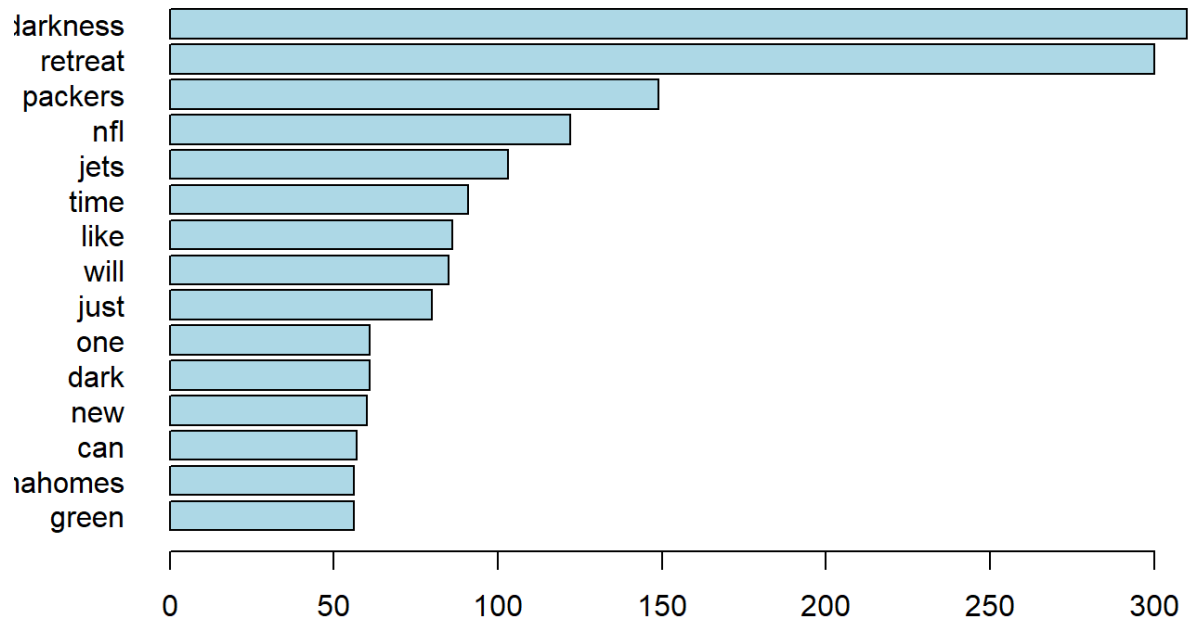
To download the tweets, I used the `search_tweets` function after I had authorized basic access for my twitter account to download tweets. I then made use of the code that was provided to create time series, maps, word-clouds, barplots of the most frequent words, cluster analysis of frequent words, a social network analysis and then finally a sentiment analysis. For conducting this analysis, I had to clean the tweets, and this was done by making use of vectors and a corpus so that I could remove special characters and as many stop words as long with filtering out Aaron Rodgers's name so these wouldn't pop up in the word clouds or word frequency. Also going through the code, I made sure to make any edits to chart titles, axis or the data visualization code itself so that it would best relate to Aaron Rodgers tweets. Once the filtering and cleaning process had been completed, I was safe to continue to gather the visualizations that I mentioned earlier. Below I included the four most important ones in my opinion that tell the most from the tweets that were downloaded. The word-cloud and word frequency help me understand more the words that are associated with Aaron Rodgers tweets, the cluster dendrogram helps me understand the way these words are used together (hence cluster) and the sentiment analysis helps me understand what are the emotions going into these tweets about Aaron Rodgers, who has become quite a controversial character in recent years around football circles.

4 Different Visualizations

Word-cloud

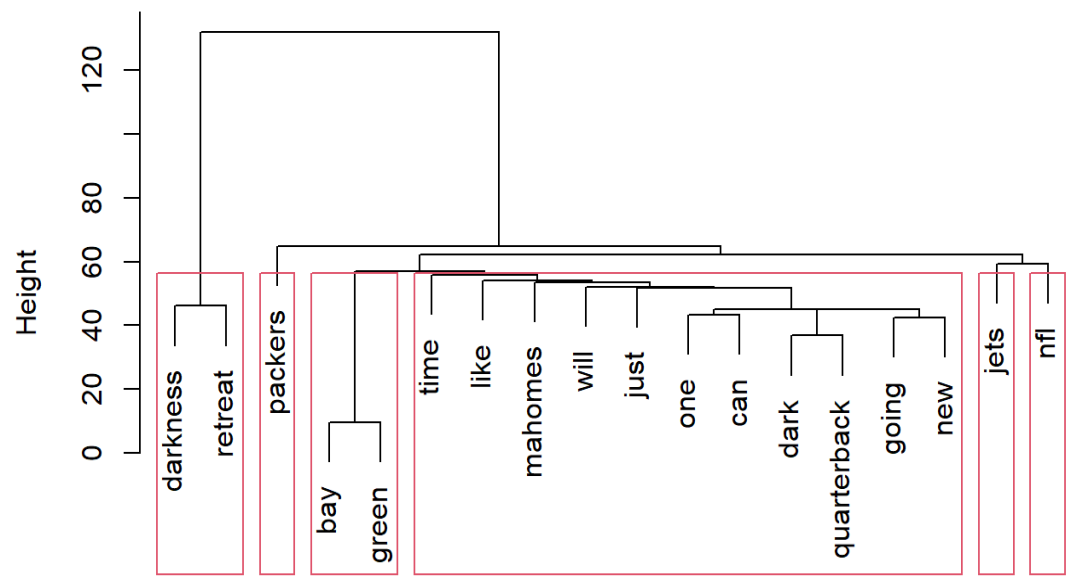


Most Frequent Words



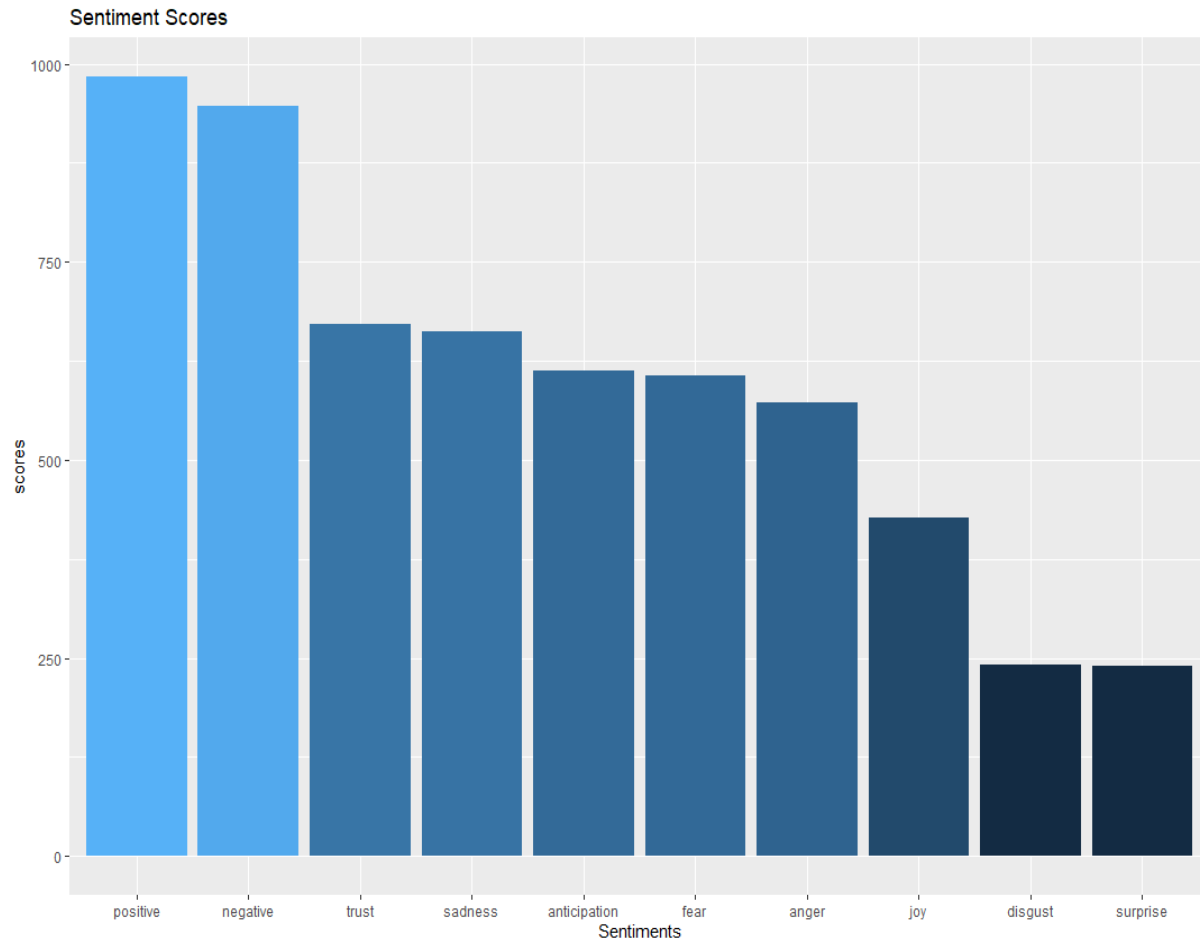
Cluster Dendrogram

Cluster Dendrogram



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Sentiment Analysis



Three Insights about Investigation

- Much more interest in Aaron Rodgers's darkness retreat visit than discussions on what possible teams he could end up on. This does make sense as Rodgers hasn't made clear what his next moves are moving forward but has said once he leaves the darkness retreat he will be making a decision shortly after. He left the retreat in Oregon a few days ago, which also explains the increase in tweets about him as the whole darkness retreat story was wacky and caught a lot of people's attention.
- Looking at the word clouds and cluster dendrogram, I see a pattern in that many of the words being used are either related to potential teams he could end up on (the Raiders, Jets and Packers in this case), his darkness retreat visit, other quarterbacks on the market (like Derek Carr, Jimmy Garoppolo or Lamar Jackson if the Ravens move on from him) and finally other popular quarterbacks currently still playing or top tier that Rodgers gets compared to.
- The Sentiment Analysis surprised me in that I wasn't expecting Positive or trust to be so high. Since Rodgers is as polarizing as he is, I would have thought Negative, Sadness and Anticipation would be the top three due to Packers fans not happy about Rodgers potentially leaving and the rest of the NFL fandom waiting to see where he goes next if he does indeed leave.

Three take-aways about Unstructured Data

- With the right functions and code in place, you can clean up unstructured data and create appealing visualizations quite easily.
- Word clouds and frequency of words be a great asset early on in trying to find patterns with unstructured data.
- When working with tweets about people's names, make sure to get rid of their first and last names along with firstnamelastname and FIRSTNAMELASTNAME along with the stop words you are trying to get rid of for the analysis.